



Question Set

Question 1

Given prices 100, 97, 101, 99, 104, 102, and 108, compute the six arithmetic returns and six log returns. Which return type aggregates exactly over time?



Question 2

Use the returns in Question 1. Compute the sample mean and sample volatility of the arithmetic returns.

Question 3

Annualize the daily volatility from Question 2 using 252 trading days. State the assumption behind this scaling.

Question 4

For an AR(1) model

$$X_t = c + \phi X_{t-1} + \epsilon_t$$

derive the unconditional mean and variance under stationarity.

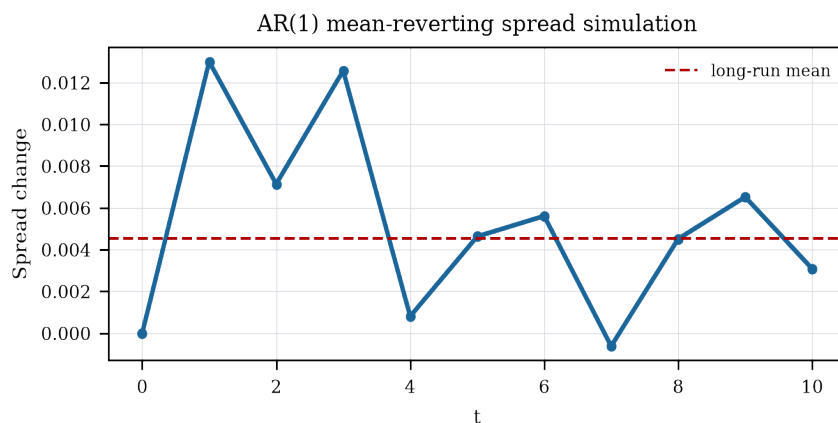
Question 5

An AR(1) spread model has

$$c = 0.001, \quad \phi = 0.78$$

and innovation volatility 1.20 percent. Compute the long-run mean, long-run volatility, and next forecast when

$$X_t = 0.030$$

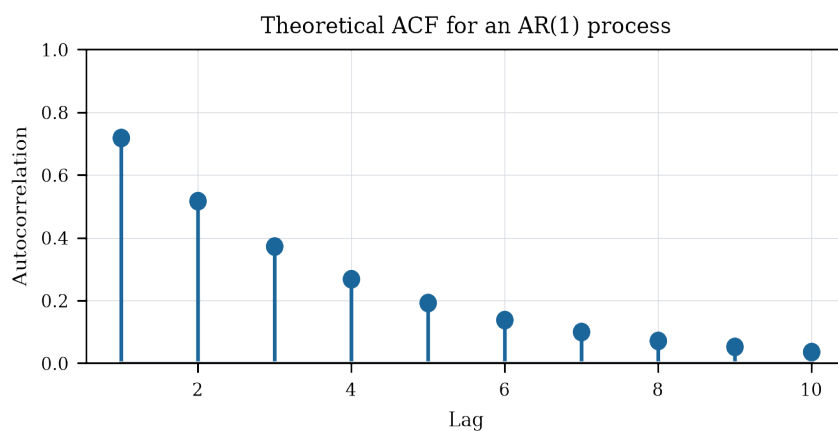


Question 6

For an AR(1) with

$$\phi = 0.72$$

compute the theoretical autocorrelations at lags 1, 3, and 10. Interpret the plotted decay.



Question 7

A time series follows

$$X_t = X_{t-1} + \epsilon_t$$

with

$$\text{Var}(\epsilon_t) = \sigma^2$$

Derive

$$\text{Var}(X_t)$$

when

$$X_0 = 0$$

Why is this a unit-root problem?

Question 8

Derive the OLS estimator for

$$y = X\beta + u$$

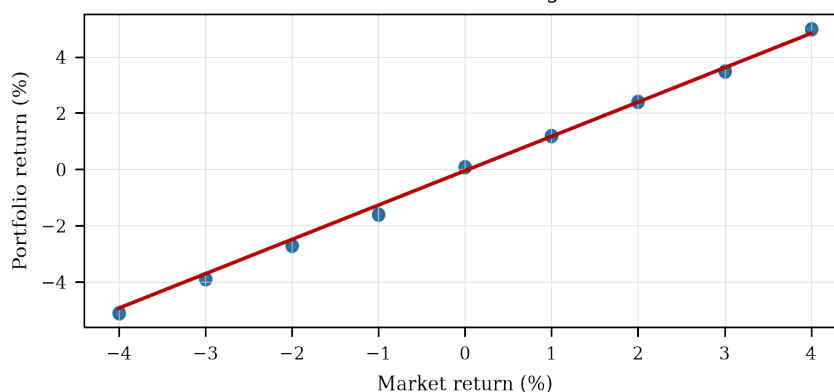
and state the exogeneity condition needed for unbiasedness.

Question 9

Using the scatter plot, estimate whether the portfolio has positive market beta. Then compute predicted portfolio return for a market return of -3 percent if

$$\hat{r}_p = -0.05\% + 1.22r_m$$

Portfolio return beta regression



Question 10

A portfolio return regression has

$$\hat{\alpha} = 0.0002, \quad \hat{\beta} = 1.4$$

market volatility 1.5 percent, and residual volatility 0.8 percent. Decompose variance into systematic and idiosyncratic parts.

Question 11

Explain omitted-variable bias in a factor-risk regression and give the one-omitted-factor formula.

Question 12

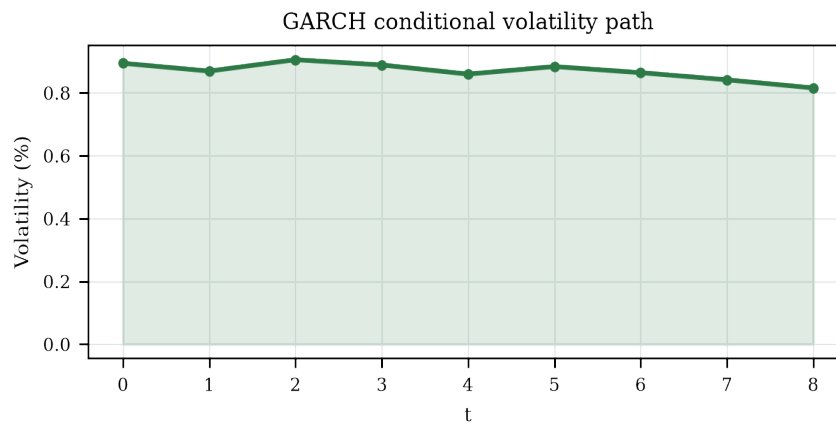
A GARCH(1,1) model is

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

For

$$\omega = 0.000002, \quad \alpha = 0.08, \quad \beta = 0.90$$

check stationarity and compute long-run volatility.



Question 13

Using the GARCH model in Question 12, compute tomorrow's conditional volatility if

$$\epsilon_t = -2\%, \quad \sigma_t = 1.1\%$$

Question 14

A 99 percent VaR model has 8 exceptions in 250 days. Compute the expected number of exceptions and give the likelihood of exactly 8 exceptions under the correct model.

Question 15

For two nonstationary price-level series, explain why a high R-squared regression can be spurious. What would you check before using the relationship?



Question 16

Define cointegration. If

$$Y_t - 1.3X_t$$

is stationary while X and Y are each

$$I(1)$$

what does that imply for relative-value risk?

Question 17

A residual series from a factor model has sample autocorrelation

$$\hat{\rho}_1 = 0.32$$

with

$$n = 100$$

Use the rough standard error

$$1/\sqrt{n}$$

to assess whether residual autocorrelation is material.

Question 18

A regression estimated on daily returns gives residuals with volatility 1 percent and clear volatility clustering. Why are usual OLS standard errors suspect, and what would you use instead?

Question 19

A 60-day EWMA volatility estimate uses

$$\lambda = 0.94$$

yesterday's variance

$$\sigma_t^2 = 0.0001$$

and today's squared return

$$r_t^2 = 0.0004$$

Compute tomorrow's variance forecast.

Question 20

A two-factor regression has estimated betas

$$\beta_{\text{EQ}} = 1.1, \quad \beta_{\text{IR}} = -4.5$$

where the rate factor is measured in decimal yield changes. Estimate PnL impact for equity return -2 percent and rate move +5 bps.